

Trainings ▾

Preparatory Steps

- Remove the intake and exhaust piping from the turbocharger.

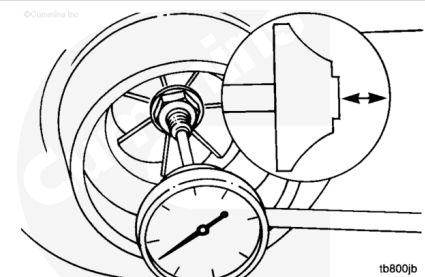


Measure

Note : Check the model of turbocharger fitted to the engine, this is for: **Holset®, Model HC5**

To measure axial clearance, use Part Number ST-537 Dial Depth Gauge or a dial indicator. Measure the axial motion.

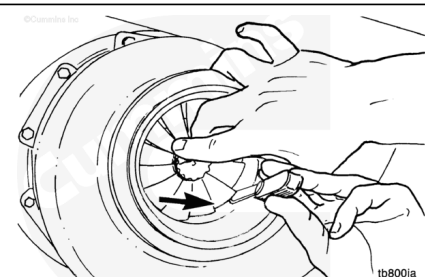
Holset® HX5 - Axial Clearance		
mm		in
0.05	MIN	0.002
0.13	MAX	0.005



If the axial clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. See Bulletin 3810243, HC5 Series Turbocharger Shop Manual, for rebuild instructions.

Use your hand pressure to push the compressor impeller end of the shaft towards the housing.

Use a wire type feeler gauge or a feeler gauge **no wider** than 13 mm [1/2 inch]. Measure the clearance between the impeller and the housing at the **minimum** clearance point.



Holset® HX5 - Compressor Impeller Radial Clearance

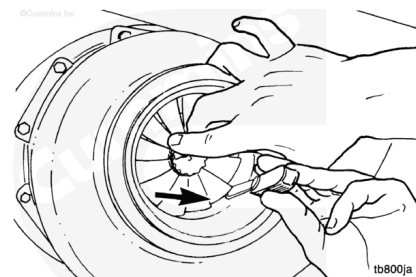
mm		in
0.15	MIN	0.006
0.46	MAX	0.018

Perform the same measurement procedure for the turbine wheel.

Holset® HX5 - Turbine Wheel Radial Clearance

mm		in
0.20	MIN	0.008
0.53	MAX	0.021

If the clearance exceeds the specification, the part **must** be replaced or rebuilt. See Bulletin 3810243, HC5 Series Turbocharger Shop Manual for rebuild instructions.



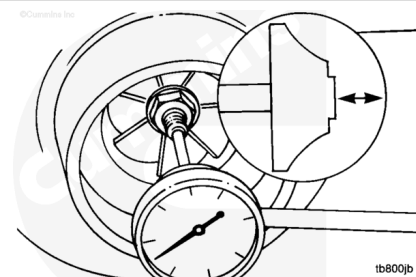
Note : Check the model of turbocharger fitted to the engine, this is for: **Brown Boveri®, Model RR-153 and RR-151**

To measure axial clearance, use Part Number ST-537 Dial Depth Gauge or a dial indicator. Measure the axial motion.

BBC (RR-151) and (RR-153) End Clearance

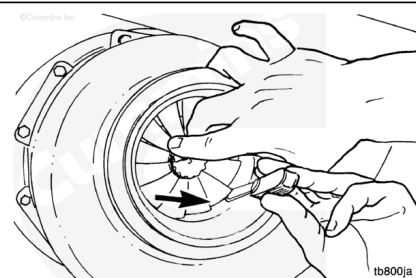
mm		in
0.09	MIN	0.004
0.13	MAX	0.005

If the end clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. See Bulletin 3810235, Exhaust - Gas Turbocharger RR-153, Instructions for Operation and Maintenance or Bulletin 3810325, Exhaust - Gas Turbocharger, RR-151-12, Instructions for Operation and Maintenance.



Note : The compressor and turbine casings do not require removal to measure the radial clearance. The parts shown are removed for clarity.

Use a feeler gauge to measure the radial clearance.



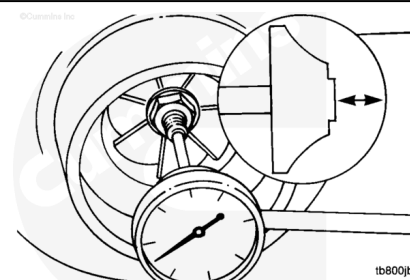
BBC (RR-151) and (RR-153) Bearing Radial Clearance				
	mm		in	
RR-153		0.55	MAX	0.021
RR-151		0.75	MAX	0.030

If the radial clearance exceeds the specification, the part **must** be replaced or rebuilt. See Bulletin 3810235, Exhaust - Gas Turbocharger, RR-153, Instructions for operation and Maintenance or Bulletin Number 3810315, Exhaust - Gas Turbocharger, RR-151-12, Instructions for Operation and Maintenance.

Note : Check the model of turbocharger fitted to the engine, this is for: AiResearch™, Model T-18A

Use a dial indicator to measure the end clearance.

AiResearch™ T-18A End Clearance		
mm		in
0.10	MIN	0.004
0.23	MAX	0.009



If the end clearance exceeds the specifications, the turbocharger **must** be replaced or rebuilt. See Bulletin 33379091, Turbocharger Component Shop Manual, for rebuild instructions.

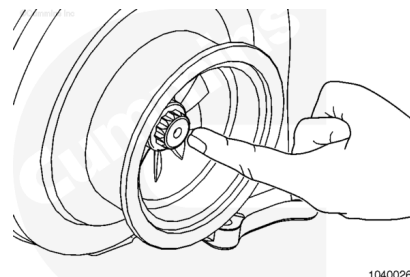
Note : Check the model of the turbocharger fitted to the engine, this is for: Honeywell®, all models.

Specifications for checking radial and axial clearance are **not** available.

Inspect for signs of contact between the turbocharger vanes and housing.

Check the bearing system for wear:

- Hold both ends of the turbocharger shaft and rotate the shaft while applying pressure down on the compressor side and up on the turbine side. It should rotate freely and there should be **no** contact between the compressor wheel and the compressor housing, or the turbine wheel and the turbine housing.

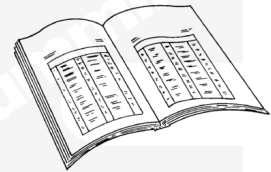


Replace the turbocharger if any problems are identified.

Finishing Steps

- Install the intake and exhaust piping to the turbocharger.

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